Last-Iterate Rates of Stochastic Fast Krasnosel'skiĭ–Mann Iterations: Momentum, Implicit Variance Reduction, and Decaying Noise

Abstract

Blue highlights the research questions and reading guide;
red highlights the main answers and limitations.

Key words. nonexpansive operator, fixed-point problem, Fast KM, stochastic oracle, almost-sure convergence.

1 Introduction

We consider the fixed-point problem

$$\text{find } x^\star \in \mathbb{R}^d \quad \text{such that} \quad Tx^\star = x^\star, \tag{FP}$$

where $T : \mathbb{R}^d \to \mathbb{R}^d$ is a nonexpansive operator, that is, $\|Tx - Ty\| \le \|x - y\|$ for all $x, y \in \mathbb{R}^d$, whose set of fixed points $\operatorname{Fix} T := \{x \in \mathbb{R}^d : Tx = x\}$ is nonempty. Throughout, only noisy evaluations of T are available.

For exact evaluations, the fixed-point residual of KM with a constant relaxation parameter satisfies $\|x_k - Tx_k\| = o(1/\sqrt{k})$ [2]. The Fast KM method of Boţ and Nguyen [2, Algorithm 2.1] improves this to

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \to x^\star \in \operatorname{Fix} T, \tag{1.1}$$

for a momentum parameter $\alpha > 2$.

With noisy evaluations, Tx_k in Fast KM is replaced by an oracle response $\widehat{T}_k(x_k) = Tx_k + e_k$. For the value at x_{k-1} there are two implementations.

- *Re-query with the same sample.* The sampled operator of iteration k is evaluated at x_k and at x_{k-1}. This uses two queries per iteration and an oracle that can evaluate one sample at two points [7, Assumption 2.3]. Variance-reduced methods build on this access [7–9].
- *Reuse.* The response $\widehat{T}_{k-1}(x_{k-1}) = Tx_{k-1} + e_{k-1}$ of the preceding step is stored and used again. This keeps one query per iteration, as in the deterministic method, and needs only fresh samples.

We thus study the reuse implementation. Table 1 raises three questions:

1. *(Q1) Last iterate of stochastic KM.* Under bounded variance, the best available last-iterate bound is $O(\log k/k^{1/6})$, whereas a randomly selected iterate attains $O(k^{-1/4})$. Does the last iterate necessarily lose this rate? (Section 3)
2. *(Q2) Implicit variance reduction in Fast KM.* Under bounded variance, does the momentum of Algorithm 1 with fixed $\alpha > 2$ reduce the variance implicitly, so that its last iterate improves on that of stochastic KM? (Section 4)
3. *(Q3) Fast KM under decaying noise.* How fast must the variance σ_k^2 decay for Algorithm 1 to attain $k\|x_k - Tx_k\| \to 0$ and $x_k \to x_\infty$ almost surely? (Section 5)

Table 1: Comparison of last-iterate residual guarantees for a nonexpansive T on $\mathbb{R}^d$.

Method	Momentum coefficient	Last-iterate rate
KM [2]	—	$o(1/\sqrt{k})$
Stochastic KM [3]	—	$O(\log k/k^{1/6})$
Fast KM [2]	$\frac{k}{k+\alpha} \to 1$	$o(1/k)$
Stochastic Fast KM, reuse (this paper)	$\frac{k}{k+\alpha} \to 1$	need not vanish in mean square (Prop. 4.2)
Same, $\mathbb{E}\sigma_k^2 \asymp k^{-q}$ (this paper)	$\frac{k}{k+\alpha} \to 1$	$o(1/k)$ a.s.: open; mean-square: $q > 2$ necessary

One oracle query per iteration. Stochastic KM is stated in expectation; the fixed-parameter counterexample concerns mean-square convergence. The last row states an almost-sure research objective; its mean-square necessary condition is a separate conclusion. Proposition 4.2 assumes $\mathbb{E}\sigma_k^2 \geq \sigma^2 > 0$.

2 Preliminaries

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{F}_k)_{k\geq 0}$ a filtration, where $\mathcal{F}_k$ contains the information available before query k. At an $\mathcal{F}_k$-measurable point x_k, the oracle returns

$$\widehat{T}_k(x_k) = Tx_k + e_k \tag{2.1}$$

with an $\mathcal{F}_{k+1}$-measurable error e_k.

Assumption 2.1 (Oracle)**.** For every $k \geq 0$,

$$\mathbb{E}[e_k \mid \mathcal{F}_k] = 0 \qquad \text{and} \qquad \sigma_k^2 := \mathbb{E}\big[\|e_k\|^2 \,\big|\, \mathcal{F}_k\big] \text{ with } \mathbb{E}\sigma_k^2 < \infty. \tag{H1}$$

3 The last iterate of stochastic KM

In this section, we review existing convergence results for stochastic Krasnosel'skiĭ–Mann iterations in [3].

The stochastic KM iteration is

$$x_{k+1} = (1-\lambda_{k+1})x_k + \lambda_{k+1}\widehat{T}_k(x_k), \qquad \widehat{T}_k(x_k) = Tx_k + e_k. \tag{3.1}$$

Under bounded variance, a randomly selected iterate generated by (3.1) attains the order $k^{-1/4}$ [3, Corollary 5.4], [4]. This leads to the following question:

(Q1) For a general nonexpansive T and bounded variance, can the last iterate of stochastic KM attain the order $k^{-1/4}$?

Let $D := \operatorname{dist}(x_0, \operatorname{Fix} T)$ and set $\tau_k := \sum_{j=1}^k \lambda_j(1-\lambda_j)$, $\varphi(y) := \min\{1, 1/\sqrt{\pi y}\}$ and

$$\nu_k^2 := \sum_{j=1}^k \big(w_j^k\big)^2 \mathbb{E}\sigma_{j-1}^2, \qquad w_j^k := \lambda_j \prod_{i=j+1}^k (1-\lambda_i),$$

We recall the conditions used for the last-iterate bound in [3].

Assumption 3.1. Either T has bounded range and $\bar{\kappa} \geq \sup_x \|Tx - x_0\|$, or $S := \sum_{j\geq 2} \lambda_j \nu_{j-1} < \infty$ and $\bar{\kappa} \geq 2D + S$.

The following theorem gives a bound on the expected fixed-point residual of the last iterate of stochastic KM [3, Theorem 2.10].

Theorem 3.1. *Let T be nonexpansive with $\operatorname{Fix} T \neq \emptyset$, and let Assumptions 2.1 and 3.1 hold. Then, for every $k \geq 1$,*

$$\mathbb{E}\|x_k - Tx_k\| \leq \bar{\kappa}\,\varphi(\tau_k) + 2\sum_{j=2}^k \lambda_j\,\varphi(\tau_k - \tau_j)\,\nu_{j-1} + 4\nu_k. \tag{3.2}$$

We now consider the special case of Assumption 2.1 in which the second moments are uniformly bounded.

Assumption 3.2 (Bounded variance)**.** There exists a finite constant $\sigma \geq 0$ such that $\sup_{k\geq 0} \mathbb{E}\sigma_k^2 \leq \sigma^2$.

Corollary 3.2. *[3, Theorem 4.4] Under the assumptions of Theorem 3.1 and Assumption 3.2, let* $\lambda_k = (k+1)^{-a}$ *with* $a \in (\frac{1}{2}, 1)$*. Then* $\tau_k \asymp k^{1-a}$ *and it holds that*

$$\mathbb{E}\|x_k - Tx_k\| \lesssim \underbrace{\frac{1}{\sqrt{\tau_k}}}_{deterministic} + \underbrace{\sigma\sqrt{k}\,\lambda_k}_{noise} = \frac{1}{k^{(1-a)/2}} + \frac{\sigma}{k^{a-1/2}}.$$

Moreover, it holds that $\mathbb{E}\|x_k - Tx_k\| = O(\log k/k^{1/6})$ *with* $a = \frac{2}{3}$.

Answer to (Q1). In the literature examined here, no: for a general nonexpansive T,

$$\mathbb{E}\|x_k - Tx_k\| = O\Big(\frac{\log k}{k^{1/6}}\Big) \quad \text{(last iterate)}, \qquad \mathbb{E}\|x_R - Tx_R\| = O\Big(\frac{1}{k^{1/4}}\Big) \quad \text{(random iterate)}.$$

The limitations are as follows:

1. For exact evaluations,

$$\|x_{k+1} - Tx_{k+1}\| \leq \|x_k - Tx_k\| \quad \Longrightarrow \quad \|x_k - Tx_k\|^2 \leq \frac{D^2}{\tau_{k+1}};$$

   with noise, $\mathbb{E}\|x_k - Tx_k\|^2$ need not be monotone, and the summed bound applies only to x_R [3, Remark 5.3].

2. *Order of the last iterate:* in (3.2), $1/\sqrt{\tau_k} \asymp \sqrt{k}\,\lambda_k \asymp k^{-1/6}$ at $a = \frac{2}{3}$ (Corollary 3.2).

4 Implicit variance reduction in Fast KM

4.1 Fast KM iteration

Boţ and Nguyen [2] accelerate the KM iteration using Nesterov's momentum updates. For fixed parameters $\alpha > 2$ and $0 < s \leq 1$, and initial points $x_0, x_1 \in \mathbb{R}^d$, their Fast KM method is

$$\begin{aligned} x_{k+1} = &\left(1 - \frac{s\alpha}{2(k+\alpha)}\right) x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)} Tx_k \\ &+ \frac{sk}{k+\alpha}(Tx_k - Tx_{k-1}), \qquad k \geq 1. \end{aligned} \tag{4.1}$$

The stepsize s can be absorbed into the operator: (4.1) applied to T coincides with (4.1) with $s = 1$ applied to $M := (1-s)\,\mathrm{Id} + sT$, which is nonexpansive with $\operatorname{Fix} M = \operatorname{Fix} T$, and an oracle for T with error e_k yields an oracle for M with error se_k. We therefore take $s = 1$ without loss of generality:

$$x_{k+1} = \left(1 - \frac{\alpha}{2(k+\alpha)}\right) x_k + \frac{\alpha}{2(k+\alpha)} Tx_k + \frac{k}{k+\alpha}(Tx_k - Tx_{k-1}). \tag{4.2}$$

4.2 The reuse implementation of Stochastic Fast KM

With noisy evaluations, Tx_k in (4.2) is replaced by a fresh response $\widehat{T}_k(x_k) = Tx_k + e_k$, while Tx_{k-1} is not re-queried: we reuse the response stored at the previous step,

$$\widehat{T}_{k-1}(x_{k-1}) = Tx_{k-1} + e_{k-1},$$

Algorithm 1 Stochastic Fast KM with reuse

Input: $x_0, x_1 \in \mathbb{R}^d$, $\alpha > 2$.
Query $\widehat{T}_0(x_0)$ and store it.
for $k = 1, 2, \dots$ **do**
  Query $\widehat{T}_k(x_k) = Tx_k + e_k$.
  $x_{k+1} = x_k - \frac{\alpha}{2(k+\alpha)}\big(x_k - \widehat{T}_k(x_k)\big) + \frac{k}{k+\alpha}\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big)$.
  Store $\widehat{T}_k(x_k)$.
end for

where e_{k-1} is the error already realized at step $k-1$. Each iteration therefore calls the oracle exactly once, and the update reads

$$x_{k+1} = x_k - \frac{\alpha}{2(k+\alpha)}\big(x_k - \widehat{T}_k(x_k)\big) + \frac{k}{k+\alpha}\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big).$$

Compared with (4.2), the update carries the error

$$\eta_k = \frac{k+\alpha/2}{k+\alpha}\, e_k - \frac{k}{k+\alpha}\, e_{k-1}, \tag{4.3}$$

$$\mathbb{E}[\eta_k \mid \mathcal{F}_k] = -\frac{k}{k+\alpha}\, e_{k-1}, \tag{4.4}$$

which is nonzero in general, since e_{k-1} is $\mathcal{F}_k$-measurable. In stochastic optimization, momentum averages the sampled gradients and reduces the variance implicitly [5, 6]. Algorithm 1 is stochastic KM with momentum and one query per iteration, which leads to the second question:

(Q2) Under bounded variance, does the momentum of Algorithm 1 with fixed $\alpha > 2$ reduce the variance implicitly, so that its last iterate improves on that of stochastic KM?

4.3 Noise propagation under reuse

Set $\rho_k := \frac{\alpha}{k+\alpha}$ for $k \geq 0$. Then $\frac{\alpha}{2(k+\alpha)} = \frac{\rho_k}{2}$ and $\frac{k}{k+\alpha} = 1 - \rho_k$, and the update of Algorithm 1 reads

$$x_{k+1} = x_k - \frac{\rho_k}{2}\big(x_k - \widehat{T}_k(x_k)\big) + (1-\rho_k)\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big).$$

Compared with the exact update, it carries the error

$$\eta_k = \Big(1 - \frac{\rho_k}{2}\Big)e_k - (1-\rho_k)\, e_{k-1}.$$

Existing analyses of Fast KM concern exact evaluations or summable perturbations [10]. Treating η_k as a summable perturbation requires additional decay assumptions, which are not implied by bounded variance.

A reduction. Following [3, Section 2.1], we reformulate the iteration with an auxiliary sequence that absorbs the errors. Let $u_0 := 0$, $u_1 := \frac{1}{2}e_0$, and define

$$u_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)u_k + \Big(1 - \frac{\rho_k}{2}\Big)e_k - (1-\rho_k)\, e_{k-1}, \qquad k \geq 1. \tag{4.5}$$

Subtracting this from the update and denoting $z_k := x_k - u_k$, we get

$$z_{k+1} = z_k - \frac{\rho_k}{2}\big(z_k - Tz_k - \delta_k\big) + (1-\rho_k)\big(Tz_k + \delta_k - Tz_{k-1} - \delta_{k-1}\big),$$

which is the deterministic Fast KM iteration with errors $\delta_k := Tx_k - Tz_k$ satisfying $\|\delta_k\| \leq \|x_k - z_k\| = \|u_k\|$.

For stochastic KM, the corresponding process is an average of the errors and converges to zero under the noise and stepsize conditions in that analysis, so the effective errors δ_k become small [3, Section 2.1]. The next lemma shows that u_k is not an average.

Lemma 4.1 (Noise propagation under reuse)**.** *Let Assumption 2.1 hold. Set* $v_k := u_k - \big(1 - \frac{\rho_{k-1}}{2}\big)e_{k-1}$ *and* $b_k := \big(1 - \frac{\rho_k}{2}\big)\big(1 - \frac{\rho_{k-1}}{2}\big) - (1 - \rho_k)$*. Then* $v_1 = 0$*,* $0 \le b_k \le \frac{\rho_k \rho_{k-1}}{4}$*, and for all* $k \ge 1$*,*

$$u_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)e_k + v_{k+1}, \qquad v_{k+1} = \Big(1 - \frac{\rho_k}{2}\Big)v_k + b_k\, e_{k-1}, \tag{4.6}$$

$$\begin{aligned} \mathbb{E}\|u_{k+1}\|^2 &= \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\sigma_k^2 + \mathbb{E}\|v_{k+1}\|^2, \\ \mathbb{E}\|v_{k+1}\|^2 &= \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\|v_k\|^2 + b_k^2\, \mathbb{E}\sigma_{k-1}^2. \end{aligned} \tag{4.7}$$

Proof. Since $\rho_0 = 1$, $v_1 = u_1 - \frac{1}{2}e_0 = 0$. Substituting $u_k = v_k + \big(1 - \frac{\rho_{k-1}}{2}\big)e_{k-1}$ into the recursion of u_k gives the recursion of v_k. A direct computation gives $b_k = \frac{\alpha(\alpha-2)}{4(k+\alpha)(k+\alpha-1)}$, which lies in $\big[0, \frac{\rho_k \rho_{k-1}}{4}\big]$. The vector v_k depends only on $e_0, \dots, e_{k-2}$, and v_{k+1} is $\mathcal{F}_k$-measurable. Conditional unbiasedness therefore removes the cross terms. □

In Lemma 4.1, the stored error e_{k-1} enters with the net coefficient $b_k = O(1/k^2)$: its two occurrences in consecutive updates cancel to leading order. The new error e_k enters with the weight

$$\underbrace{\frac{\rho_k}{2}}_{\text{relaxation}} + \underbrace{(1 - \rho_k)}_{\text{momentum}} = 1 - \frac{\rho_k}{2} \;\to\; 1,$$

because it appears both in the relaxation step and in the momentum term. If $\inf_k \mathbb{E}\sigma_k^2 > 0$, then u_k does not converge to zero in mean square. The estimate $\|\delta_k\| \le \|u_k\|$ therefore does not establish that the effective errors vanish in mean square.

Proposition 4.2. *Let* $T = 0$ *and assume* $\mathbb{E}\sigma_k^2 \ge \sigma^2 > 0$ *for all* k*. Then the iterates of Algorithm 1 satisfy, for every* $k \ge 1$*,*

$$\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \ge \frac{\sigma^2}{4}.$$

Proof. For $T = 0$, the update reads $x_{k+1} = \big(1 - \frac{\rho_k}{2}\big)x_k + \big(1 - \frac{\rho_k}{2}\big)e_k - (1 - \rho_k)e_{k-1}$. Since x_k and e_{k-1} are $\mathcal{F}_k$-measurable, conditional unbiasedness gives $\mathbb{E}\|x_{k+1}\|^2 \ge \big(1 - \frac{\rho_k}{2}\big)^2 \mathbb{E}\sigma_k^2 \ge \frac{\sigma^2}{4}$, because $\rho_k \le 1$. □

Remark 4.3. Bravo and Cominetti observe that stochastic KM carries implicitly a form of variance reduction that operates as the iterates evolve [3, Remark 2.3]. For Algorithm 1, Lemma 4.1 shows that this variance reduction acts only on the stored errors. The new error enters through the momentum term, whose coefficient $1 - \rho_k$ tends to one for the parameters of Fast KM. For comparison, momentum-based methods control the contribution of new stochastic information through tunable parameters [5, 6].

Answer to (Q2). For fixed $\alpha > 2$: no. The residual of Algorithm 1 need not even vanish in mean square under bounded variance (Proposition 4.2).

5 Fast KM under decaying noise

Proposition 4.2 shows that, even for $T = 0$, persistent noise can prevent mean-square convergence of the residual. We therefore study decaying noise and ask:

> *(Q3) How fast must the variance* σ_k^2 *decay for Algorithm 1 to attain* $k\|x_k - Tx_k\| \to 0$ *and* $x_k \to x_\infty$ *almost surely?*

For deterministic variance profiles $\sigma_k^2 \asymp k^{-q}$, the proof of Proposition 4.2 gives

$$\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \ge \Big(1 - \frac{\rho_k}{2}\Big)^2 \mathbb{E}\sigma_k^2 \ge \frac{1}{4}\mathbb{E}\sigma_k^2 \qquad (T = 0).$$

Thus the root-mean-square rate $(\mathbb{E}\|x_k - Tx_k\|^2)^{1/2} = o(1/k)$ requires $q > 2$, while (N6-ℓ^2) holds if and only if $q > 3$. The former is a necessary condition for a mean-square rate; the latter is a candidate sufficient condition for almost-sure convergence and rates, whose applicability to Algorithm 1 remains to be established.

The objective is to identify noise conditions under which

$$k\|x_k - Tx_k\| \longrightarrow 0 \qquad \text{almost surely,} \tag{5.1}$$

and, for a Fix$T$-valued random variable x_∞,

$$x_k \to x_\infty \qquad \text{almost surely.} \tag{5.2}$$

The candidate condition is the decaying-noise condition of [1, Theorem 3.1]: $\sum_k s_k^2 t_k^2 \sigma_k^2 < \infty$ or $\sum_k s_k t_k \sigma_k < \infty$ almost surely, where s_k is the stepsize and $t_k := 1 + \sum_{i \geq k} \prod_{j=k}^{i} \alpha_j$ for the extrapolation coefficients α_k. For Algorithm 1, $s_k \equiv 1$ and $\alpha_k = 1 - \rho_k$, so $t_k = \frac{k+\alpha-1}{\alpha-1}$ and the condition reads

$$\text{(N6-}\ell^2\text{)} \ \sum_k (k+1)^2 \sigma_k^2 < \infty \qquad \text{or} \qquad \text{(N6-}\ell^1\text{)} \ \sum_k (k+1)\sigma_k < \infty \tag{5.3}$$

almost surely. For deterministic profiles $\sigma_k^2 \asymp k^{-q}$, (N6-ℓ^2) holds if and only if $q > 3$, and (N6-ℓ^1) if and only if $q > 4$.

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